

Driving Process' Analysis and HUD Design Based on Conditional Autonomous Traffic Safety

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Abstract. With the rapid increasing of car quantity, traffic safety problem has become more and more serious. Traffic crash is also a major world public health problem; hence, it has become a topic of interest among many scholars. This study intends to analyze the factors that affect driving safety under different driving scenarios. The results of this study are based on conditional autonomous driving. The context-aware conditional autonomous safety driving process and the relationships among three elements, namely, the environment, the driver and the car, are analyzed based on the Haddon matrix and the system attribution model. The vehicle used is a conditional autonomous car capable of context awareness; it can send useful information as feedback to the driver and interact with driver behavior. Afterward, the results are applied in parking scenario and lane changing scenario, and the factors that influence safety during parking and lane changing are analyzed. Then, the obtained information is validated based on driver perspective, and a design concept for head-up display is proposed. This design is expected to assist drivers in parking and lane changing without accident.

Keywords: Conditional autonomous · Traffic safety · Driving process · HUD

1 Introduction

With the continuous development of society, cars have become the most common transportation in our daily life. With the rapid increasing of car quantity, traffic safety problem has become more and more serious. Traffic crash is a major world public health problem [1]; hence, it has become a topic of interest among many scholars. An increasing number of vehicles are currently equipped with advanced driver assistance systems (ADAS), which improve car autonomy in terms of specific driving functions [2]. Automation is a very popular topic in the field of automobile and SAE International's On-Road Automated Vehicle Standards Committee has defined six levels of driving automation [3]: 0 (no automation), 1 (driver assistance), 2 (partial automation), 3 (conditional automation), 4 (high automation), and 5 (full automation). Among them, in 0–2 levels, human driver monitors the driving environment and in 3–5 levels, automated driving system monitors the driving environment. In 3 (conditional automation) level, system executes the steering and acceleration/deceleration and monitors the driving environment. Human driver performs when getting the dynamic

driving task. According to BASt level, 3 level is equivalent to highly automated and to NHTSA level, it is equivalent to 3 level. Considering practical applications, this study chooses 3 (conditional automation) level as the research object.

In the field of safety driving, many theories, some of which are universal, have been applied. For example, scholars have proposed driver models to describe driver performance. Many vehicle driving models are based on either cybernetics or control theory, cognitive psychology as information processing or a mix thereof [4]. Explaining how control can be achieved in a driving situation has been the main objective of driver models [5]. The contextual control model (COCOM) by Hollnagel in 1993 provides a framework for examining control in different contexts, whereas the extended control model (ECOM) in 2003 aims to describe driving control at high levels [4]. These models share the single driver perspective, or in some cases, the driver-vehicle system, in approaching the task of driving. Another theory or model that aims to explain the process of driving is the accident-causing theory [5]. This theory is an accident model that corresponds to the main component of the principium of safety and is an important part of accident theory. It temporarily avoids the concrete characteristics of hazard sources and contents as well as accident modes. Instead, it abstractly considers the person, machine, goods and environment in the system under study, and consequently, identifies accident causes, processes, and results in nature.

Although many methods and models have been presented in the field of safety driving, no method has yet been established for analyzing the process of conditional autonomous driving. This study intends to bridge this research gap. It proposes the process of conditional autonomous driving and the relationships among three elements based on several theories, including the Haddon matrix [6]. A conditional autonomous car is capable of context-awareness; thus, it can send useful information to driver and interact with driver behavior. Finally, the findings of this study are applied in a parking scenario, and a design concept for head-up display (HUD) is formulated based on the results.

2 Analysis of Context-Aware Conditional Autonomous Driving Process

Two theories are used to analyze the conditional autonomous driving process. The first theory is the Haddon matrix [6], which is the most commonly used paradigm in the injury prevention field. This matrix considers factors related to personal, vector or agent, and environmental attributes before, during, and after an injury or death. The relative importance of different factors and design interventions can be evaluated using this framework. We propose three elements involved once the driver take action, namely, the driver, the environment and the car, based on the Haddon matrix. The second theory is system attribution model, which is the last stage of accident model development. This model theoretically shows the importance of multiple-factor interaction in accident evolution. It presents the notion that the human factor is not the only consideration that must be analyzed in the context of entire system in a driving

scenario. Thus, this study analyzes both the relationships among the three elements and process of individual driving behavior based on semi-autonomous driving.

A driving scenario is highly complex, and driving behavior is influenced by many factors. Therefore, this study presents certain assumptions as follows.

- The scenario is simple. This research focuses on a single scenario with a specific task, such as parking or lane changing.
- The object is unique. Only one car or one driver is present in a particular scenario, and other cars and drivers are included in the environment. Communication between drivers or cars is disregarded, and only communication among environment, driver, and car is considered.
- The driving scenario is safe. Only the appropriate processes and normal relationships are considered.
- The vehicle is conditional automation. It quantifies and obtains the environmental and driver information. The vehicle can determine which information the driver requires and provide such data after filtering.
- The driver is willing to comply with the recommendations of the car.

We analyze the processes and relationships among the three elements by considering these prerequisites (Fig. 1).

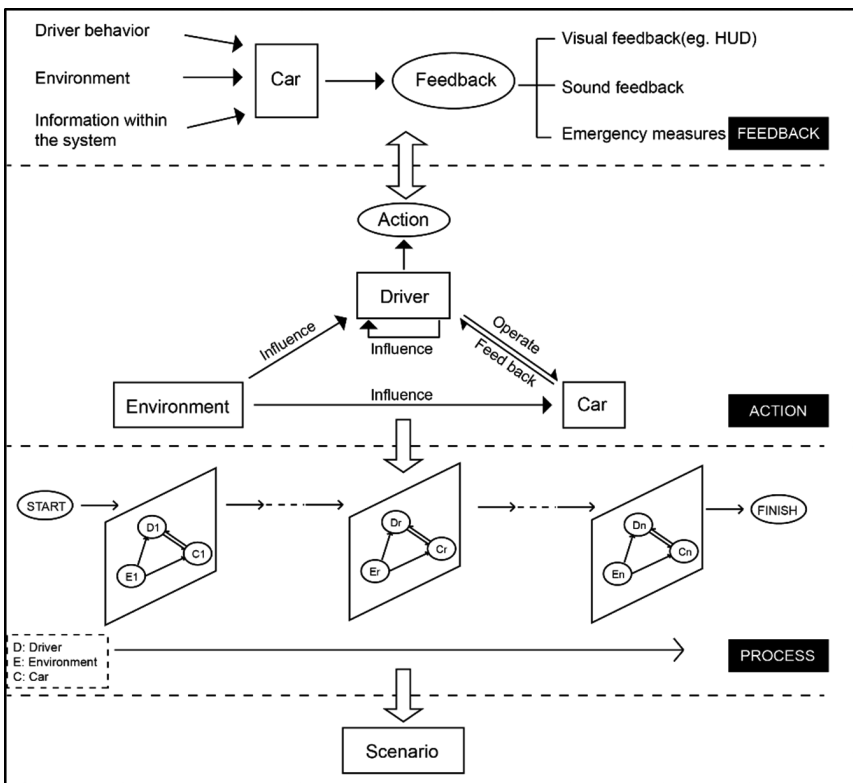


Fig. 1. Driving process and relationships among environment, driver, and car

Figure 1 includes three aspects of information as follows.

- **Driving process:** The driving process is composed of several step-by-step actions. Each scenario involves its own task and process and requires several actions to achieve the task. The order of such actions influences driving safety.
- **Relationships among the environment, the driver, and the car:** The environment can influence both the driver and the car. The driver operates the car, whereas the car provides feedback to the driver. The driver can also influence himself/herself. Both the basic information and conditions of the driver can affect his/her behavior. Environmental factors can be roughly divided into internal and external factors. The internal factors can then be divided into passenger diversity, electronic devices (e.g., mobile phone), internal environment (e.g., air quality and seat size), and car feedback. External factors can be mainly divided into five categories: road conditions, motor vehicle conditions, nonmotor vehicle conditions, road signs, and natural environment.
- **Car feedback:** The vehicle is conditional automation. Feedback is multiple. It includes visual feedback, sound feedback, and emergency measures. Feedback and driver's actions are interactional. Car's feedback can also be included in the internal environmental factors.

3 Research on Parking Scenario

In this section, we choose the parking scenario as an example and use the aforementioned result for analysis. Then, we formulate a design concept for HUD [7] that assists drivers during parking. Parking situations include back parking, parallel parking, and oblique parking. Parallel parking is a general and complex parking situation. Thus, we choose parallel parking as an example for analysis.

3.1 Parallel Parking Process

The parallel parking process can be divided into three stages (Fig. 2): preparation, parking and tail-in work. In the first stage, the main work involves selecting a parking area and preparing to park. During parking, the driver switches on the turn signal, changes gears, and turns the steering wheel several times. During this stage, the driver should pay attention to the surroundings. If other vehicles and pedestrians are passing by, then the driver should avoid scraping them. During the final stage (tail-in work), the driver should turn off the turn signal and change to parking gear. If the driver cannot park in the selected area in a single attempt, then he/she should adjust until he/she has parked completely. After analyzing the parallel parking process, we identify factors that can influence the success rate of parking. These factors include the position and size of the parking area, the start position of reverse, the timing of gear change, the timing of turning the steering wheel, and the distance from other vehicles.

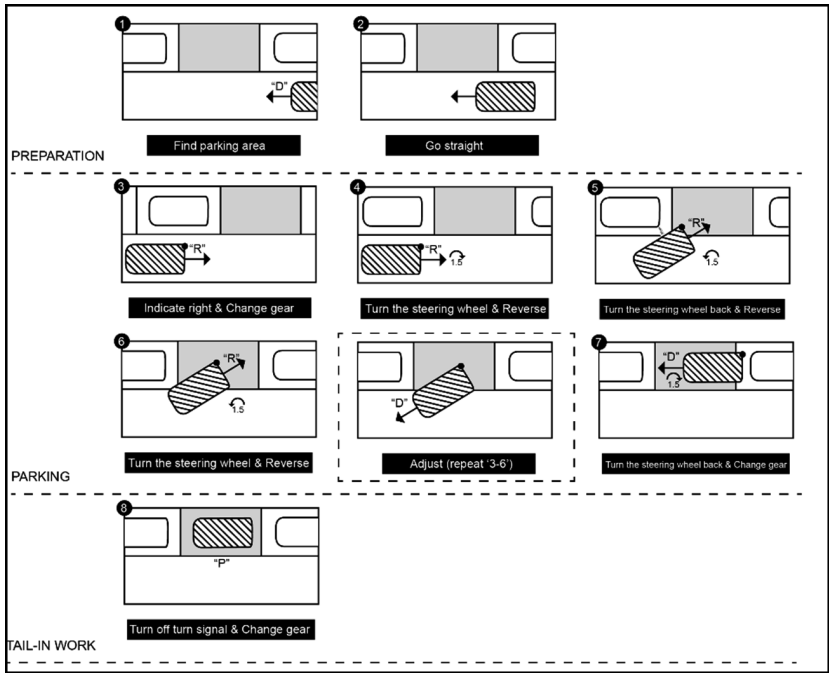


Fig. 2. The process of parallel parking

3.2 Relationships Among the Environment, the Driver, and the Car

In this section, we analyze environmental and driver factors that influence safety during parking. The environmental factors in a parking scenario can be divided into five types: road conditions, motor vehicle conditions, pedestrians, road signs, and natural environment. In this study, road conditions refer to whether the road is crowded or open. Motor vehicle conditions mainly correspond to the presence of any passing vehicle. Road signs comprise parking signs as well as the size and location of the parking area. Natural environment includes poor weather conditions and lighting in an outdoor parking lot. For a parking garage, the natural environment is the main source of light. As mentioned earlier, driver factors can be divided into basic information and driver conditions. Considering the parallel parking scenario, we propose specific factors related to drivers, namely, parking skills, driver confidence, and the ability to estimate vehicle body size. Parking skills include knowledge of the correct parking process and control of speed and distance. Driver confidence is based on the age, experience, and other basic information of the driver.

3.3 Car Feedback

In a parking scenario, feedback from the car can be divided into visual feedback on HUD, sound warning, and emergency measures. The information shown on HUD is

based on driver demand. After analysis, we find the position where the driver stops to reserve and how the steering wheel should be turned are important to a driver during parking. Finally, we conclude that the essential information consists of the parking route, key point, real-time position, next operation, gear, positions of the steering wheel and the wheel, and revised parking route. Sound feedback includes lane-departure warning, vehicle distance warning and voice reminders. Emergency measures include emergency brake and protection measures for both drivers and passengers in case of collisions.

Based on Fig. 1, we organize all the information above and conclude them in Fig. 3.

Figure 3 includes the parallel parking process, the relationships among environment, driver, and car and the car's feedback during parallel parking.

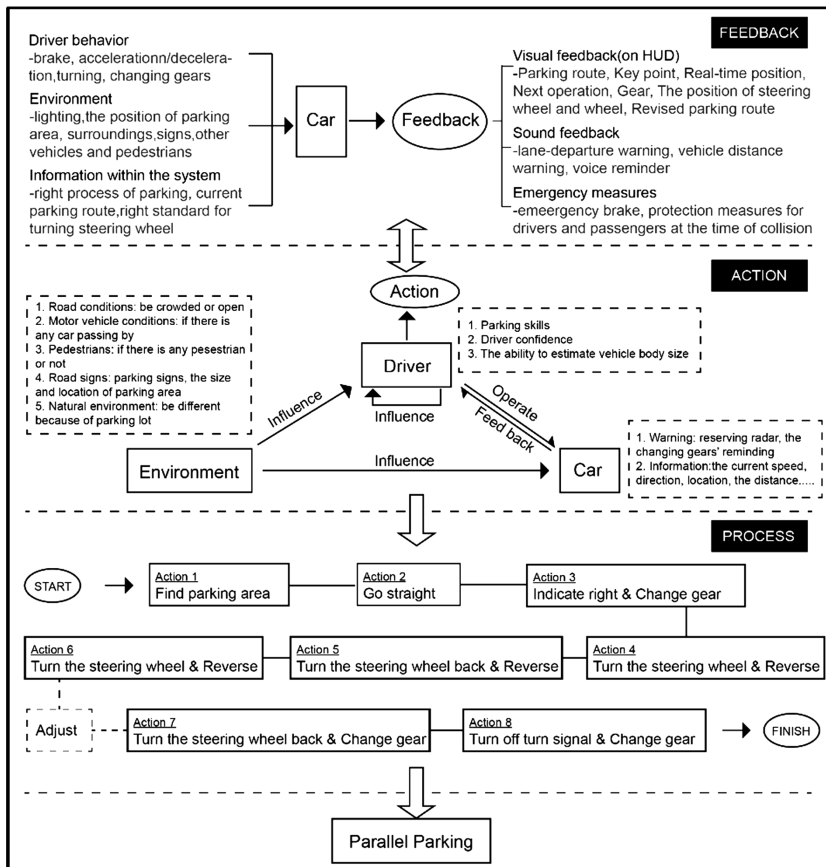


Fig. 3. Analysis of parallel parking

3.4 Design on HUD

We validate the obtained information and select HUD as the platform. We adopt the windshield as a display platform using the shadow casting technique. Hence, we can display the information on the windshield. The driver only has to watch the windshield to improve safety while driving. We perform a questionnaire survey to confirm whether the previously cited requirements are actually those needed by the driver and to sort the importance of information. The number of participants is 19. The result of questionnaire is showed in Table 1.

Table 1. Result of questionnaire

Question	The corresponding information	Probability
Don't know where and when to stop and start reversing	Parking route Point location	16/19
Don't know how much should the steering wheel turn	Steering wheel	14/19
Sometimes forget the wheels' direction	Wheel	12/19
Can't open the door because of the distance	Distance	10/19
Collision because can't see the whole situation behind the car	Rear view camera	8/19
Can't park in the area directly	Corrective route	17/19

The result of the survey (Table 1) shows that 84.2% of the participants are not aware of where and when to stop and start reversing. Approximately 73.7% of the participants do not know the number of turns that should be applied to the steering wheel. Approximately 63.2% of the participants admit to forgetting the direction of the wheels and are unable to park in a single attempt. Considering these results, we obtain the order of information as follows: parking route (key point, revised parking route, real-time position) > next operation (manner of gear change and steering wheel turn) > the position of wheel = the gear = steering wheel position.

According to the result of survey, we make sure the information that drivers need when parallel parking. We design a prototype (Fig. 4) based on the importance of information and the drivers' habit of looking at windshield.

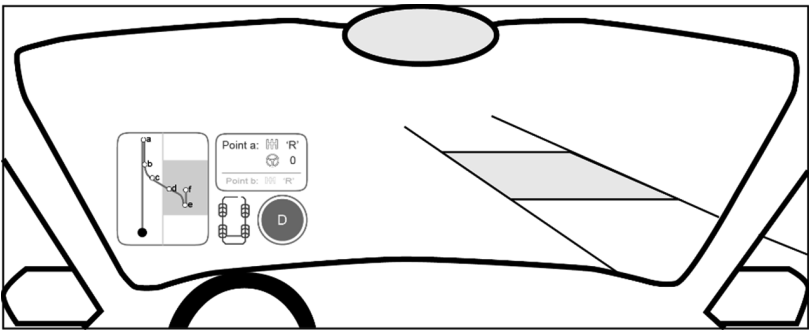


Fig. 4. HUD prototype in parking scenario

The main information in prototype (Fig. 4) includes the parking route, the point location, the steering wheel, the wheel and the distance between car and point. When the driver turns “R” gear, the information will be displayed in RVC. If drivers can not park in one time, here will be the corrective route displayed on HUD.

4 Research on Lane Changing Scenario

4.1 Lane Changing Process

In the process of driving, the frequency of lane changing scenario is high, but also more likely to cause traffic accidents scene. Therefore, it is very meaningful to design the assistance device for the safe driving behavior of lane changing. Depending on the weather and the brightness differences, we found that the driver's lane changes were more difficult due to the poor visibility at night, rainy days and haze days. However, there is a common lane changing regardless of the environment, as shown in Fig. 5. First of all is the normal driving. Secondly, according to the lane will be changing the vehicle speed and vehicle distance to determine whether changing lane. If it is suitable for lane changing, then open the lane changing lights. According to vehicle speed inside line, adjust themselves speed, acceleration or deceleration, or uniform, until a lane changing safety area appeared. At this point, immediately turn the steering wheel, change to the target lane. If changing lane succeed, turn back to the lights. If failed, return to the original lane immediately, and to find the next changing opportunity.

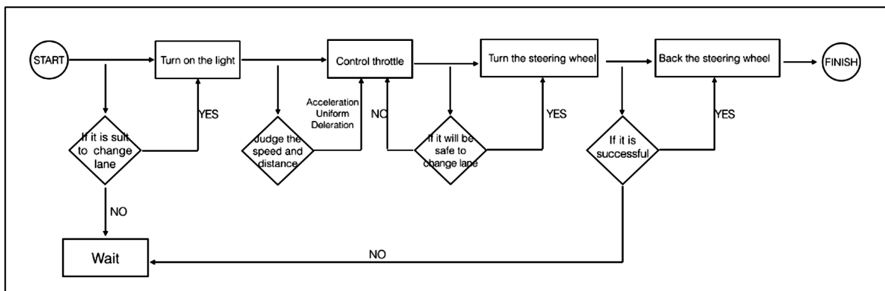


Fig. 5. The process of lane changing

As the vehicles in different situations of lane changing, the drivers' throttle control is not the same. In normal daytime weather conditions, there are five main situations for lane changing (Fig. 6):

- The right front vehicle speed is slower, take the action like acceleration or uniform. And maintain a safe distance with after car then change lane.
- The right front of the vehicle speed is faster, take the action like deceleration. And maintain a safe distance with after car then change lane.
- The right rear of the vehicle speed is faster, take the action like deceleration until overtake that car. And maintain a safe distance with after car then change lane.

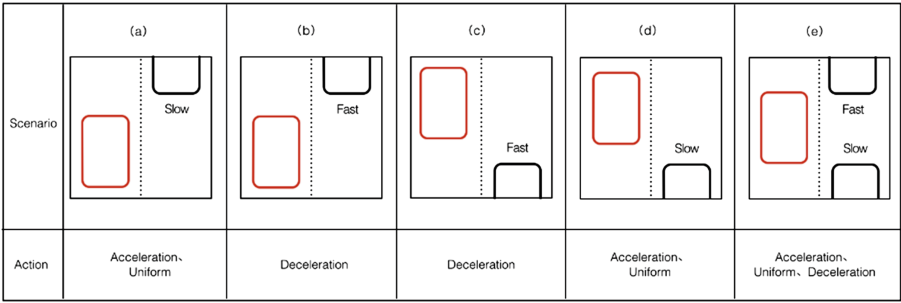


Fig. 6. Five main situations for lane changing

- (d) The right rear of the vehicle speed is slower, take the action like acceleration or uniform until overtake that car. And maintain a safe distance with after car then change lane.
- (e) The right rear of the vehicle speed is slower, take the action like acceleration, uniform or deceleration. And maintain a safe distance with before and after cars then change lane

4.2 Relationships Among the Environment, the Driver, and the Car

Under normal circumstances, environmental factors that affect driving safety can be divided into five categories: road conditions, motor vehicle conditions, pedestrians, road signs and the natural environment. In the lane changing scenario, road conditions, motor vehicle conditions, road signs and the natural environment are a greater impact factors on drivers' driving behavior and safety. The road conditions, mainly refers to whether the road is crowded and the type of road. The more vehicles on the road, the more crowded, the greater difficulty of changing lanes, and will affect the changing lane speed; type of road is mainly having impact of vehicle speed, such as highway lane changing, the driver's speed can relatively high. Motor vehicle mainly refers to the speed of other vehicles, and their distance from the car and the upcoming driving behavior. Here, the other vehicle refers to a vehicle in the lane in which the driver wants to change the lane and a vehicle that wants to change lane to the current lane. Road signs will mainly affect whether the driver is suitable for lane changing and whether the lane should be changed, in addition to lane does not allow lane change, overtaking, if the driver wants to turn right at the junction must change lanes to allow the right lane. Natural environment mainly includes two aspects, one day/night, on the other hand is bad weather, such as rain, haze days, snow days and so on, these natural environments will affect the driving environment and the driver's line of sight, in different environments, the driver needs to set aside the safety lane change will not be the same. Pedestrians of this factor in the lane scene of the impact of the other four factors are not large, lane changing scenario generally occur at a certain speed on the road, and pedestrian contact opportunities less, so pedestrians in this scenario factors can be ignored.

In lane changing scenario, the drivers' personal factors will also affect the safety of vehicles, mainly including the following aspects: the driver's personal skills, personal driving habits and personal basic information. Personal skills mainly refer to the driver of the other vehicle speed and vehicle distance of the ability to judge; personal habits mainly refer to the process of lane changing to see mirror habit, turning lights time, the speed of the habit, etc. Personal basic information, including the driver's character and driving experience, timid or self-confidence, driving experience much or little will affect the safety of lane change process.

Finally, the factors on the vehicle itself, in the lane changing scenario mainly refers to the speed of their vehicles, their speed will affect the relative speed of other vehicles, and in different speeds, lane changing safety distance will be different, these determine the appropriate time to lane change, and ultimately affect the lane change security.

4.3 Car Feedback

In lane change scenario, the feedback information to the driver after obtaining the relevant information about the driver and the environment can be divided into three main aspects: visual feedback (HUD display), sound feedback and emergency measures. Visual feedback mainly presents information that requires the driver to determine whether or not it is suitable for lane changes by personal judgment. Compared with the subjective judgment of the driver, after obtaining the information of the vehicle itself, the driver and the environment, the vehicle can show more rational information after the calculation and screening, which can help the driver avoid the safety accident caused by the mistake of subjective judgment. After analysis, the final decision on the information presented in the HUD has the following parts: the current vehicle speed, the proposed driving behavior (acceleration/constant speed/deceleration), the current environment is suitable for lane (whether with other vehicles to maintain a safe lane distance), real-time vehicle conditions within the safety range (if other vehicles appear? What's situation of their speed and distance?), and turn signal lights.

There are two types of sound feedback, voice prompts and alarm tones. Under normal circumstances, in order to avoid distracting the driver's attention, we will minimize the voice prompts to simple and necessary alarm sound as an adjunct to help the driver in the driving process to avoid dangerous situations. In the course of the lane change, the sound feedback appears mainly in two situations, the vehicle distance detection and the lane departure detection, when the vehicle is too close or the vehicle deviates from the lane there will appear an alarm tone.

The final emergency measures are not very different in different scenarios. This feature mainly considers the extreme case where the car's pre-set protection system replaces the driver's behavior in extreme danger, as well as the driver and the car in a dangerous situation Protection of passengers within the measures.

Based on Fig. 1, we organize all the information above and conclude them in Fig. 7.

Figure 7 includes the lane changing process, the relationships among environment, driver, and car and the car's feedback during lane changing.

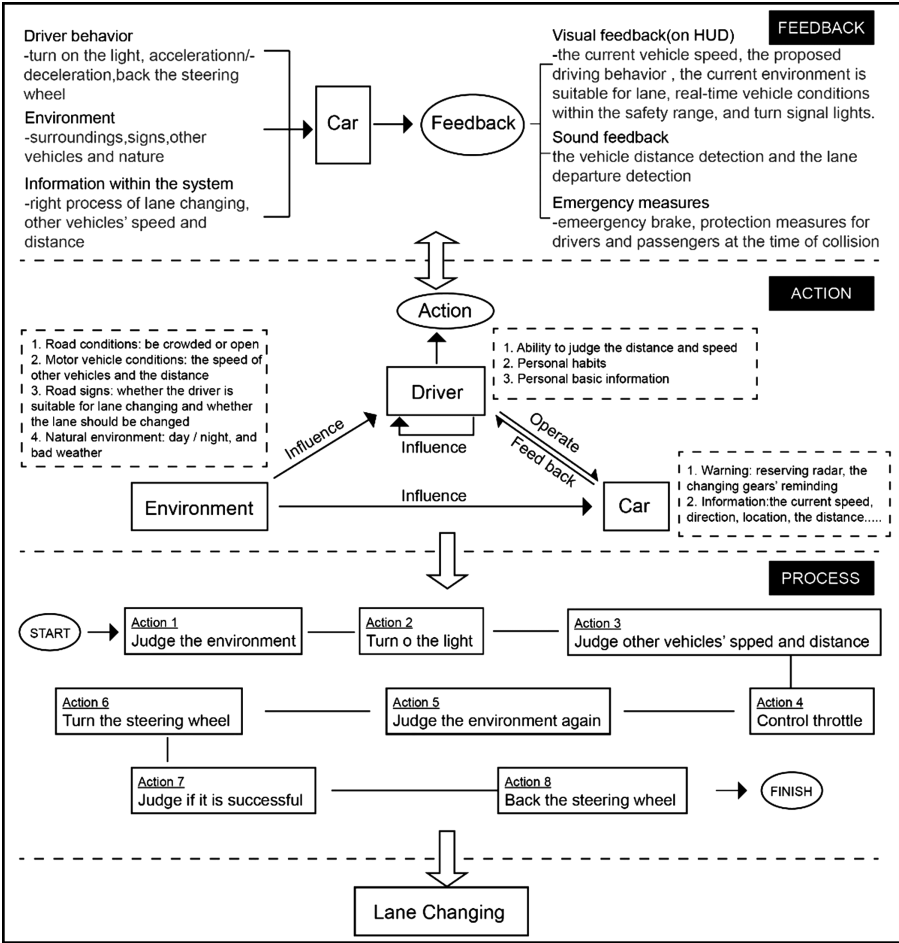


Fig. 7. Analysis of lane changing

4.4 Design

According to the result of analysis, we make sure the information that drivers need when lane changing. We design a prototype (Fig. 8) based on the importance of information and the drivers' habit of looking at windshield.

The main information in prototype (Fig. 8) includes the current vehicle speed, the proposed driving behavior (acceleration/constant speed/deceleration), the current environment is suitable for lane, real-time vehicle conditions within the safety range, and turn signal lights.

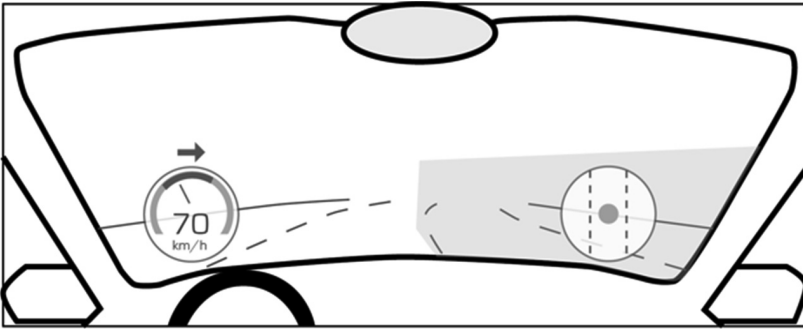


Fig. 8. HUD prototype in lane changing scenario

5 Conclusion and Future Work

This paper aims to analyze the process of conditional autonomous driving and the relationships among elements based on context-aware safety driving. We propose three elements involved in a single action, namely, the environment, the driver, and the car, based on the Haddon matrix. We analyze the relationships among the three elements and the process involved in a single driving scenario using the system attribution model, which is the last stage in accident model development. The driving is highly complex, and driving behavior is influenced by many factors. Hence, we initially make certain assumptions, such as the involvement of a simple scenario, a unique object, a safe driving scenario, a semi-autonomous car, and a compliant driver who follows the recommendations of the car. Considering these prerequisites, we propose the process involved and the relationships among the environment, the driver, and the car in a single driving scenario. To adopted car is conditional automation and capable of context awareness, and thus, it can send useful information as feedback to the driver and interact with driver behavior. Then, we choose parallel parking scenario and lane changing scenario to apply the obtained results. We analyze the parallel parking process and lane changing process, environmental and driver factors that influence safety driving, and car feedback. Finally, we develop a design concept for a parking and lane changing assistance device on HUD.

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